Savannah L. Ferretti

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EDUCATION

Ph.D. in Earth System Science, University of California, Irvine — Jun 2026
Make it Rain—Understanding Thermodynamic Controls on South Asian Monsoon Rainfall

M.S. in Earth System Science, University of California, Irvine — Dec 2023

B.S. in Earth & Atmospheric Sciences, Cornell University — Dec 2020

EXPERIENCE

Postdoctoral Researcher — Nov 2026–Present
Kerner Group, Arizona State University — Tempe, AZ

Postdoctoral Researcher — Aug–Oct 2026
Pritchard & Baldwin Groups, University of California, Irvine — Irvine, CA

- Advanced ongoing research on South Asian monsoon rainfall variability, focusing on manuscript development and publication

Graduate Student Researcher — Sep 2021–Jun 2026
Pritchard & Baldwin Labs, University of California, Irvine — Irvine, CA

- Used data-driven and physics-based approaches to identify & explain drivers of South Asian monsoon rainfall variability
- Developed an interpretable machine learning methodology that used neural networks & equation discovery to learn physically meaningful relationships from high-dimensional atmospheric data
- Built open-source Python pipelines on GitHub for cloud data access, large-array processing, uncertainty analysis, model evaluation, & visualization in HPC environments
- Communicated findings through peer-reviewed publications & conference presentations

Consulting Analyst — Mar–Aug 2021
Higher Education Research Services, Huron Consulting Group — New York, NY

- Built Excel-based financial planning & reporting tools for a research nonprofit
- Developed long-term resource forecasts & presented recommendations to executive stakeholders

Undergraduate Research Assistant — Jun–Aug 2020
Commane Lab, Lamont-Doherty Earth Observatory — Palisades, NY

- Quantified changes in atmospheric methane concentrations during the NYC COVID-19 shutdown
- Applied meteorological analysis & transport modeling to attribute emissions to likely source regions

Student Administrative Assistant — Jan–Dec 2020
CALS Office of Admissions, Cornell University — Ithaca, NY

- Increased prospective student yield through creation of an interactive ArcGIS campus tour
- Improved student engagement through execution of an office-wide social media strategy

Undergraduate Research Assistant Sep 2019–May 2020
Pritchard Lab, Cornell University Ithaca, NY

- Processed satellite imagery to construct time series of volcanic thermal anomalies
- Investigated pre-eruptive thermal signals for inclusion in NASA's AVTOD volcano database

SKILLS

Python (PyTorch, PySR, Dask, Xarray, Cartopy, Matplotlib) • GitHub • High-Performance Computing • NetCDF/HDF5/Zarr • Bash • Linux • Microsoft Excel (Certified Specialist) • LaTeX • WordPress

PUBLICATIONS

S. L. Ferretti, J. W. Baldwin, M. S. Pritchard, & T. Beucler. Discovering equations for the thermodynamics of South Asian monsoon precipitation. Submitted to the *Journal of Advances in Modeling Earth Systems*.

S. L. Ferretti, J. Lin, S. Shamekh, J. W. Baldwin, M. S. Pritchard, & T. Beucler. (2026). Data-driven integration kernels for interpretable nonlocal operator learning. *Environmental Data Science*, *5*, e31.

S. L. Ferretti, M. S. Pritchard, F. Ahmed, L. Peng, & J. W. Baldwin. (2025). Explaining South Asian monsoon rainfall seasonality using a metric of plume buoyancy. *Geophysical Research Letters*, *52*, e2025GL115546.

L. Peng, P. N. Blossey, W. M. Hannah, …**S. L. Ferretti**, …& M. S. Pritchard. (2025). Resolving low cloud feedbacks globally with E3SM High-Res MMF: Agreement with LES but stronger shortwave effects. *Journal of Advances in Modeling Earth Systems*, *17*, e2025MS005003.

S. Yu, Z. Hu, A. Subramaniam, …**S. L. Ferretti**, …& M. S. Pritchard. (2025). ClimSim-Online: A large multi-scale dataset and framework for hybrid ML–physics climate emulation. *Journal of Machine Learning Research*, *26*, 1-85.

S. Yu, W. M. Hannah, L. Peng, …**S. L. Ferretti**, …& M. S. Pritchard. (2023). ClimSim: A large multi-scale dataset for hybrid physics–ML climate emulation. *NeurIPS 2023 Datasets & Benchmarks Track*, **Outstanding Paper Award**.

A. M. Jenney, **S. L. Ferretti**, & M. S. Pritchard. (2023). Vertical resolution impacts explicit simulation of deep convection. *Journal of Advances in Modeling Earth Systems*, *15*, e2022MS003444.

PRESENTATIONS

(Oral) **S. L. Ferretti**, J. Lin, S, Shamekh, …& T. Beucler (2026). Data-Driven Integration Kernels for Interpretable Nonlocal Operator Learning. *Climate Informatics 2026*. Lausanne, VD, CH.

(Invited Oral) **S. L. Ferretti**, J. Lin, S, Shamekh, …& T. Beucler (2025). Making Data-Driven Climate Prediction Interpretable by Design. *Learning the Earth with Artificial Intelligence and Physics Lectures in Climate Data Science*. Virtual.

(Oral) **S. L. Ferretti**, T. Beucler, M. S. Pritchard, …& J. W. Baldwin (2025). Learning Nonlocal Controls on South Asian Monsoon Rainfall. *American Geophysical Union Fall Meeting*. New Orleans, LA, USA.

(Poster) **S. L. Ferretti**, M. S. Pritchard, & J. W. Baldwin (2025). Towards Data-Driven Discovery of Thermodynamic Controls on South Asian Monsoon Rainfall. *Learning the Earth with Artificial Intelligence and Physics Year 5 Meeting*. Briarcliff Manor, NY, USA.

(Poster) **S. L. Ferretti**, L. Peng, J. W. Baldwin, & M. S. Pritchard (2024). Spatiotemporal Stress-Testing of a Process-Oriented Diagnostic for Rainfall. *American Geophysical Union Fall Meeting*. Washington DC, USA.

(Poster) **S. L. Ferretti**, J. W. Baldwin, N. Liu, …& M. S. Pritchard (2024). Unraveling CMIP6 Biases in Summer Monsoon Rainfall Over the Arabian Sea and Western India. *104th American Meteorological Society Annual Meeting*. Baltimore, MD, USA.

(Poster) **S. L. Ferretti**, N. Liu, J. W. Baldwin, & M. S. Pritchard (2022). Understanding the Monthly Variation in the Upstream Enhancement of Indian Summer Monsoon Precipitation Near the Western Ghats. *American Geophysical Union Fall Meeting*. Chicago, IL, USA.

(Invited Oral) **S. L. Ferretti** (2022). Using Models to Understand Climate Change. *Making Modern Science*, University of California, Irvine. Irvine, CA, USA.

(Poster) **S. L. Ferretti**, B. Dalton, L. D. Schiferl, …& R. Commane (2020). Assessing the Meteorological Impact on Methane (CH_4) Emission Changes in New York City During the COVID-19 Shutdown. *American Geophysical Union Fall Meeting*. Virtual.

AWARDS & FELLOWSHIPS

Advancing Inclusive Excellence Award — Jan–Jun 2023
School of Physical Sciences, University of California, Irvine — $ 400

Diversity, Equity, & Inclusion Graduate Leaders Fellowship — Sep 2022–Jun 2023
School of Physical Sciences, University of California, Irvine — $ 24,720

Class Award — Apr 2018
Cornell University Class Council

TEACHING

University of California, Irvine, Introduction to Earth System Science — Jun–Jul 2026
Teaching Assistant — Virtual

University of California, Irvine, Air Quality Management — Apr–Jun 2024
Teaching Assistant — Irvine, CA

University of California, Irvine, Earth System Chemistry — Sep–Dec 2023
Teaching Assistant — Irvine, CA

University of California, Irvine, Diversity in STEM Seminar — Sep–Dec 2022
Coordinator — Irvine, CA

SERVICE & ENGAGEMENT

Journal of Geophysical Research Atmospheres, Reviewer — Nov 2025–Present

UCI ESS First-Year Mentoring Program, Mentor — Sep 2022–Aug 2024

UCI Community Geoscience Initiative, Fellowship Application Reviewer — Jun 2023

UCI CLEWS Climate-Tech Seminar Series, Panelist — Jun 2023

UCI ESS Inclusive Excellence Committee, Organizer — Sep 2022–Jun 2023

NCAR Community Earth System Model Tutorial, Participant — Aug 2022

AGU Bridge Program New Student Orientation, Panelist — Jul 2022

Seed Consultant Group, Volunteer Project Consultant — Mar–May 2022

GREEN Program Nepal, Microgrid Systems for Rural Development, Participant — Jan 2020

Cornell University Dance Team, Cofounder & Vice President — Oct 2017–Dec 2020

Cornell University College Mentors for Kids, Mentor — Aug 2017–May 2018

RELEVANT COURSEWORK

GRADUATE

Machine Learning for Environmental Engineering & Science • Atmospheric Dynamics • Geoscience Modeling & Data Analysis • Environmental Fluid Mechanics & Turbulence • Global Physical Climatology • Humans in the Earth System • Planning for Climate Justice • Communication Skills for Environmental Scholars

UNDERGRADUATE

Atmospheric Thermodynamics & Hydrostatics • Climate Dynamics • Microclimatology • Meteorological Observations & Instrumentation • Synoptic Meteorology • Evolution of the Earth System • Interior of the Earth • Oral Communication • Leading People & Effective Communication